

Retrieval-Augmented Generation (RAG) Architecture Guide

A Technical Breakdown of Modular Ingestion and Generation Pipelines

Overview

Retrieval-Augmented Generation (RAG) is a design pattern that enhances the capabilities of Large Language Models (LLMs) by grounding their responses in external, vetted data sources. Rather than relying solely on the static weights of the model, RAG dynamically retrieves relevant text chunks or multimodal elements from a vector store, assembles them into a contextual prompt, and feeds them into the generator. This minimizes hallucinations and ensures domain-specific accuracy.

Key Components

A production-ready RAG pipeline consists of the following phases:

- **1. Document Ingestion:** Files (PDF, MD, CSV, etc.) are parsed, structured, and chunked into manageable snippets.
- **2. Vector Database:** Text snippets are converted to dense vector embeddings and stored in indexes (e.g. Qdrant) for fast semantic search.
- **3. Query Encoder:** User queries are embedded using the same vector space to allow similarity matching.
- **4. LLM Generator:** The query and top retrieved chunks are merged into a contextual prompt to synthesize a grounded response.

Architecture Diagram

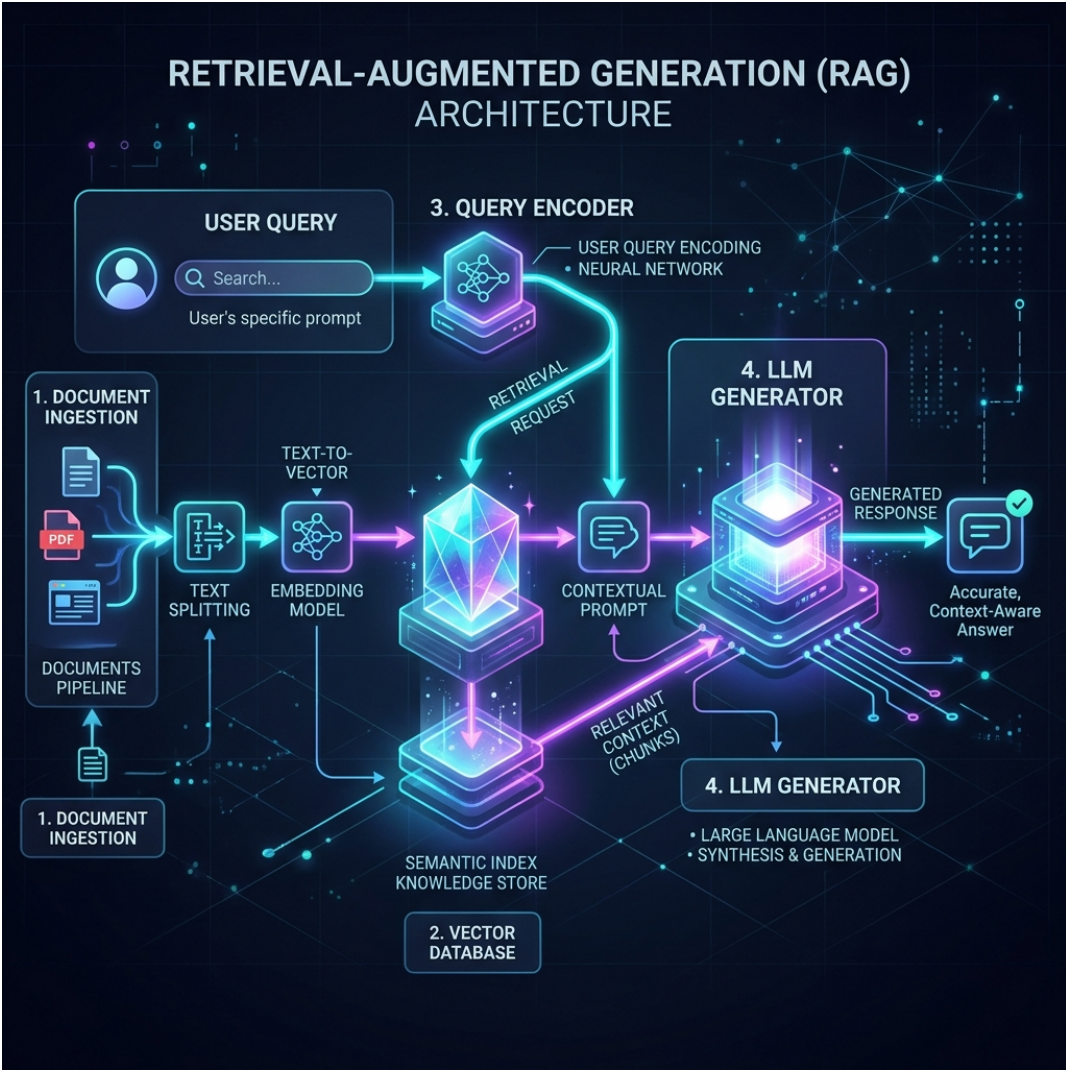


Figure 1: Conceptual Architecture of a Modern Multimodal RAG Pipeline.